• **BUCK/JERK—WHILE DRIVING IN ANY FUEL MODE—VEHICLES EQUIPPED WITH 5.4L BI-FUEL ENGINE ONLY** • **DRIVEABILITY—BUCK/JERK—WHILE DRIVING IN ANY FUEL MODE—VEHICLES EQUIPPED WITH 5.4L BI-FUEL ENGINE ONLY** • **HESITATION—WHILE DRIVING IN ANY FUEL MODE—VEHICLES EQUIPPED WITH 5.4L BI-FUEL ENGINE ONLY**	**Article No. 00-23-2**

FORD: 1999 F-250 LD
1999-2000 F-150

ISSUE

Some vehicles may exhibit an intermittent hesitation or buck while driving in any fuel mode (LPG, CNG or gasoline). This may be caused by a fault in the Alternative Fuel Sensors wire harness.

ACTION

Perform diagnostics and if necessary, replace the Alternative Fuel Sensors wire harness. Refer to the following Service Procedure for details.

The procedure below first electrically disconnects and by-passes the Alternative Fuel System to determine if the buck and jerk condition exists in the gasoline only mode. If the condition clears, this verifies a fault in the Alternative Fuel Sensors harness. The harness is then replaced.

SERVICE PROCEDURE

1. With ignition key in the Off position, disconnect the vehicle battery.
2. Disconnect lock and remove the following wires from male connector C120 (refer to Wiring Diagrams Workshop Manual information for connector cavity location):
   - Black wire from cavity 35
   - Dark blue wire from cavity 28
   - Gray wire from cavity 22
   - Gray/red wire from cavity 18
   - Gray/white wire from cavity 11
3. Cover wires removed from connector C120 and secure out of the way.
4. Disconnect connector C4202 (marked J202 on harness).
5. Remove all the wires from the male connector C4202.
6. Connect the following wires removed from the male connector C4202 into the male connector C120 (refer to Wiring Diagrams Workshop Manual information for connector cavity locations):
   - Black wire into cavity 35
   - Dark blue wire into cavity 28
   - Gray wire into cavity 22
   - Gray/red wire into cavity 18
   - Gray/white wire into cavity 11
7. Reinstall lock and connect C120.
8. With ignition key in the Off position, connect the vehicle battery.
9. Switch vehicle to run on gasoline by placing the fuel selector switch in the GAS position.
10. Retest vehicle for hesitation and buck condition.
11. If the condition does not clear, repair base vehicle, then return the wiring to its original configuration using the following steps:
    a. With ignition key in the Off position, disconnect the vehicle battery.

Copyright © 2000 Ford Motor Company

b. Disconnect lock and remove the following wires from male connector C120 (check EVTM Workshop information for connector cavity locations):
   - Black wire from cavity 35
   - Dark blue wire from cavity 28
   - Gray wire from cavity 22
   - Gray/red wire from cavity 18
   - Gray/white wire from cavity 11

c. Reconnect the wires removed from C120 into the male connector C4202, match cavity location and wire color with the female connector C4202 (marked J202 on harness).

d. Reconnect connector C4202 (marked J202 on harness).

e. Reconnect the following wires that have been covered and secured out of the way (harness is marked J202) into the male connector C120 (check EVTM Workshop information for connector cavity locations):
   - Black wire into cavity 35
   - Dark blue wire into cavity 28
   - Gray wire into cavity 22
   - Gray/red wire into cavity 18
   - Gray/white wire into cavity 11

NOTE
WHEN INSTALLING A NEW ALTERNATIVE FUEL SENSORS HARNESS UNDER STEP 12 BELOW, USE COLOR-CODED WIRES INDICATED FROM HARNESS CONNECTOR MARKED J202.

f. Reinstall lock and connect C120.

NOTE
WHEN INSTALLING A NEW ALTERNATIVE FUEL SENSORS HARNESS UNDER STEP 12 BELOW, CONNECT ALL REMAINING CONNECTORS BEFORE PROCEEDING WITH STEP g.

g. With the ignition key in the Off position, connect the vehicle battery.

h. Switch vehicle to run on LPG or CNG by placing the fuel selector switch in the ALT position.

NOTE
WHEN REPLACING THE ALTERNATIVE FUEL SENSORS HARNESS, THE KEY MUST BE IN THE OFF POSITION BEFORE CONNECTING OR DISCONNECTING THE VEHICLE BATTERY.

12. If the condition clears, replace the Alternative Fuel Sensors harness using procedures shown in Step 11 and then for the remaining portion of the harness, replace wire for wire and connector for connector. 1999 model year F-150/250 bi-fuel LPG and CNG vehicles, use part number (XL3Z-14A411-GA). 2000 model year F-150 bi-fuel LPG and CNG vehicles, use part number (YL3Z-14A411-AA). Replace wire straps as necessary.

PART NUMBER	PART NAME
XL3Z-14A411-GA	Alternative Fuel Sensors Wire Harness (1999 F-150/250 Bi-Fuel LPG/CNG Truck)
YL3Z-14A411-AA	Alternative Fuel Sensors Wire Harness (2000 F-150 Bi-Fuel LPG/CNG Truck)

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
002302A	Replace Alternative Fuel Sensor Wire Harness (Includes Time To Perform Diagnostics)	2.2 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
14A411	42

OASIS CODES: 203000, 607000, 608000, 609000, 610000, 611000, 612000, 614000